

Notes: Gan (JFE, 2007)

Collateral, Debt Capacity, and Corporate Investment: Evidence from a Natural Experiment

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1 Big picture

1.1 One-sentence summary

This note summarizes Gan (2007), which uses the Japanese land-price collapse as a natural experiment to estimate how exogenous losses in collateral value reduced manufacturing firms' debt capacity and investment, and then verifies the mechanism using matched bank-firm lending data that show the same banks cut lending more to firms with larger collateral losses.

1.2 Research question

The paper asks whether the market value of collateral matters for real corporate decisions. More specifically, it asks:

- Does a negative shock to collateral value reduce corporate investment?
- Does this effect operate through a reduction in bank debt capacity?
- Is the collateral channel separate from the lending channel generated by weakened bank balance sheets?

1.3 Core economic mechanism

The paper studies the following chain:

Land-price collapse $\Rightarrow$ Lower value of pledgeable collateral $\Rightarrow$ Lower debt capacity $\Rightarrow$ Lower investment

The key empirical challenge is that collateral is usually endogenous: firms buy assets when they invest, and collateral may be correlated with firm quality. Gan solves this by using pre-shock landholdings in Japan as the cross-sectional exposure to a large, external collapse in land prices.

1.4 Main findings

- Firms with larger pre-shock landholdings suffered greater collateral losses after the land-price collapse.
- These firms reduced investment more during 1994–1998.
- The magnitude is economically large: for an average firm, a 10 percent decline in land value reduces the investment rate by about 0.8 percentage point.
- Matched bank-firm loan data show the mechanism directly: firms with larger collateral losses are less likely to maintain lending relationships and, conditional on continuation, receive less bank credit.
- The collateral channel remains after controlling for bank health, bank real estate exposure, bank capital ratios, and bank fixed effects.

2 Incremental contribution of the paper

2.1 Contribution relative to the financial constraints literature

Earlier empirical work often inferred financial constraints from investment-cash flow sensitivity. That approach is controversial because cash flow can proxy for investment opportunities, managerial

agency problems, or expected profitability. Gan contributes by shifting the empirical object from cash flow to collateral value. Since collateral directly affects pledgeable borrowing capacity, it is closer to a primitive source of external financing frictions.

2.2 Contribution relative to the collateral-channel literature

Theoretical work such as Bernanke and Gertler (1989, 1990) and Kiyotaki and Moore (1997) emphasizes collateral amplification, but clean firm-level evidence was limited. Gan provides direct evidence using:

- a large exogenous collateral shock from Japan’s land-price collapse;
- firm-level investment data;
- matched bank-firm loan data;
- bank fixed effects to compare different borrowers of the same bank.

The last point is especially important: the paper does not only show that landholding firms invest less; it also shows that banks reduce lending more to those same firms.

2.3 Contribution relative to Japanese lost-decade explanations

The paper connects Japan’s asset-market collapse to real-sector investment through a firm collateral channel. This complements other explanations such as bank balance-sheet deterioration, evergreening, productivity slowdown, and weak aggregate demand. The paper’s central message is that the asset-price collapse hit both sides of the credit market: banks’ balance sheets and borrowers’ collateral capacity.

3 Institutional background and natural experiment

3.1 Why Japan is a useful setting

Japan in the early 1990s is an unusually clean setting because land prices collapsed dramatically after a large asset-price boom. Corporate borrowing in Japan was traditionally collateralized by land, and banks commonly relied on secured lending. Thus, a collapse in land prices directly reduced firms’ pledgeable collateral.

3.2 Timing of the shock

The empirical design uses the following timing:

- **Pre-shock exposure:** firm landholdings measured in 1989.
- **Shock:** sharp collapse in Japanese land prices from 1991 to 1993.
- **Post-shock outcomes:** investment and lending outcomes mainly during 1994–1998.

The time gap helps the paper argue that 1989 landholdings are unlikely to be chosen in anticipation of firm-specific investment opportunities during 1994–1998.

3.3 Why pre-shock landholdings identify collateral losses

Let L_i^{1989} denote firm i 's market value of land before the shock. If land prices fall by a common proportional amount $\Delta P/P < 0$, the firm's collateral loss is approximately

$$\Delta C_i \approx \frac{\Delta P}{P} L_i^{1989}.$$

Because $\Delta P/P$ is common and external to any individual firm, cross-sectional variation in L_i^{1989} identifies variation in collateral damage.

4 A simple derivation of the collateral channel

4.1 Borrowing constraint

Consider a firm with investment I_i and internal funds A_i . It can borrow B_i from banks, but borrowing is limited by collateral:

$$B_i \leq \theta C_i,$$

where C_i is pledgeable collateral value and $\theta \in (0, 1]$ is the loan-to-collateral capacity parameter.

The investment financing constraint is

$$I_i \leq A_i + B_i.$$

If the firm is financially constrained, the borrowing constraint binds, so

$$I_i = A_i + \theta C_i.$$

A collateral shock then has the direct effect

$$\frac{\partial I_i}{\partial C_i} = \theta > 0.$$

A fall in collateral value lowers investment one-for-one up to the pledgeability parameter.

4.2 Collateral damage from land-price collapse

Let collateral be land value, $C_i = P^{land} L_i$. A land-price shock changes collateral by

$$\Delta C_i = L_i \Delta P^{land}.$$

If the firm's pre-shock land quantity or land value is predetermined, then

$$\Delta I_i \approx \theta L_i^{1989} \Delta P^{land}.$$

Therefore, firms with higher L_i^{1989} should cut investment more after the price collapse, provided the borrowing constraint binds.

4.3 Reduced-form implication

The paper's reduced-form test is equivalent to estimating whether

$$\frac{I_i}{K_i} = \alpha + \delta \frac{Land_i^{1989}}{K_i} + \text{controls} + \varepsilon_i$$

has $\delta < 0$ after the collapse. The negative sign arises because higher pre-shock landholdings imply larger collateral losses when land prices fall.

4.4 Mechanism implication for bank lending

If investment falls because of collateral-induced debt capacity losses, then the same exposure should also predict weaker bank lending:

$$\Delta Loans_{ij} = \alpha_j + \delta_L \frac{Land_i^{1989}}{K_i} + X'_i \beta + R'_{ij} \gamma + u_{ij},$$

where α_j is a bank fixed effect. The key prediction is $\delta_L < 0$: within the same bank, borrowers with larger collateral losses should receive less credit.

5 Data and measurement

5.1 Firm-level data

The firm data come from the Development Bank of Japan database, which reports detailed accounting information for publicly traded nonfinancial firms. The main analysis focuses on publicly traded Japanese manufacturing firms after the collapse, primarily over 1994–1998.

5.2 Investment and capital stock

Investment is fixed investment scaled by the beginning-of-period replacement cost of capital:

$$\frac{I}{K}.$$

The replacement cost of capital is constructed using a perpetual inventory method separately for different categories of capital goods. This is important because the paper does not simply use book assets as the capital denominator.

5.3 Land value

The market value of land is also constructed using a perpetual inventory approach. The key exposure variable is

$$\frac{Land_i^{1989}}{K_i},$$

where land is measured before the shock. This variable is interpreted as exposure to collateral damage during the collapse.

5.4 Matched bank-firm lending data

A distinctive contribution of the paper is the use of matched bank-firm loan data. The lending data allow the paper to ask whether, for the same bank, firms with larger land-related collateral losses experience lower loan growth or relationship survival.

6 Empirical strategy

6.1 Investment regression

The baseline investment equation is

$$\frac{I_i}{K_i} = a + bq_i + c \frac{CashFlow_i}{K_i} + d \frac{Land_i^{1989}}{K_i} + e \left(\frac{CashFlow_i}{K_i} \times LandCo_i \right) + \text{Industry FE} + \varepsilon_i.$$

The coefficient of interest is d . Since land prices fell, a larger value of $Land_i^{1989}/K_i$ means a larger collateral-value loss. The collateral-channel prediction is

$$d < 0.$$

6.2 Why LAD instead of OLS

The investment rate is heavily right-skewed. The paper therefore estimates median or least absolute distance (LAD) regressions. This reduces the influence of extreme investment observations and makes the estimates more robust to skewness.

6.3 Bank lending regression

For the lending mechanism, the paper estimates

$$Lending_{ij} = a + bX_i + cR_{ij} + d\frac{Land_i^{1989}}{K_i} + u_j + \varepsilon_{ij},$$

where X_i includes firm controls, R_{ij} includes relationship characteristics, and u_j is a bank fixed effect. Bank fixed effects are central because they compare different firms borrowing from the same bank.

6.4 Selection model for relationship survival

The lending analysis also uses a Heckman two-stage model. The first stage models whether a bank-firm lending relationship survives, and the second stage models loan growth conditional on survival. This matters because collateral losses may affect both the extensive margin of lending relationships and the intensive margin of loan amounts.

7 Identification logic

7.1 Main identifying assumption

The identifying assumption is that, conditional on controls and industry effects, pre-shock landholdings are not systematically correlated with post-shock firm-specific investment opportunities. Formally, the causal interpretation requires

$$\mathbb{E}[\varepsilon_i \mid Land_i^{1989}/K_i, X_i, \text{Industry FE}] = 0.$$

7.2 Why the assumption is plausible

The assumption is plausible for three reasons:

- the shock to land prices is aggregate and very large;
- landholdings are measured before the shock;
- outcomes are measured several years after the pre-shock landholding variable.

The paper also controls for pre-shock investment, firm quality proxies, leverage, recent land purchases, and access to bond markets.

7.3 Threats and paper responses

- **Landholding proxies for low quality.** The paper adds controls for firm size, sales growth, profitability, and pre-shock investment.
- **Landholding firms were overinvesting before the shock.** The paper tests whether low- q firms drive the results and examines major investments.
- **Banks lending to landholding firms were weaker.** The paper uses bank fixed effects and controls for bank real estate exposure and capital ratios.
- **Regional land-price shocks vary.** The paper checks robustness using prefecture-level land-price variation.

8 Main empirical results

8.1 Investment effects

The baseline result is that pre-shock landholdings have a negative and statistically significant effect on post-shock investment. This supports the prediction that firms losing more collateral reduced investment more.

Interpretation: The coefficient on $Land/K^{pre}$ is negative in the investment regressions. Since land prices collapsed, more pre-shock land means more collateral damage. Thus, the negative coefficient means that the larger the collateral damage, the lower the subsequent investment rate.

8.2 Economic magnitude

The paper reports that a firm at the 75th percentile of landholding relative to capital has an investment rate about 1.6 percentage points lower than an otherwise similar median-landholding firm. Relative to a median investment rate of about 9 percent, this is an economically large reduction. The paper also reports that for an average firm, every 10 percent decline in land value reduces the investment rate by roughly 0.8 percentage point.

8.3 Bank-lending mechanism

The matched lending data show that collateral losses reduce debt capacity. Firms with greater collateral losses are less likely to maintain lending relationships and receive smaller loan amounts. This result is stronger than a simple investment regression because it documents the financing channel directly.

8.4 Collateral channel versus lending channel

The paper further shows that the collateral channel is not merely a proxy for bank weakness. After controlling for top-bank real estate exposure, bank capital ratios, and bank dummies, the collateral coefficient remains negative and significant. Thus, borrower collateral damage and lender balance-sheet damage are distinct channels.

9 Interpretation of key tables from the paper

9.1 Table 1: Summary statistics and Table 2: Collateral effects on fixed investments: basic results

Table 1

Summary statistics

This table presents descriptive statistics on various firm characteristics in Japan. Column (1) presents the whole-sample summary statistics averaged over the sample period between 1994 and 1998. Columns (2) and (3) present industry-adjusted statistics (adjusted by the median) for the subsamples of landholding companies and the control group. The table presents the mean of each characteristic, with the median in parentheses. Landholding companies are defined as the companies with market value of land to total replacement cost of capital above the industry top quartile. The remaining firms serve as the control group. Significance levels for the difference between the landholding companies and the control groups are based on two-tailed tests; significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	Statistics adjusted by the industry median		
	Raw statistics	Landholding companies	The control group
	The whole sample (1)	(2)	(3)
Sales (\$mil)	165.47 (48.62)	37.58*** (−5.79)***	149.95 (15.04)
Cash flow/ K	0.08 (0.03)	0.17 (−0.01)	0.00 (−0.01)
Cash/ K	1.23 (0.55)	1.94** (0.26)***	0.24 (−0.03)
EBDIT/total assets	0.03 (0.03)	−0.01* (0.00)	0.00 (0.00)
I/K	0.24 (0.09)	0.46 (0.01)**	0.07 (0.02)
Debt/total assets	0.12 (0.11)	0.00** (−0.01)***	0.02 (−0.01)
Bank debt/debt	0.56 (0.59)	−0.02*** (0.03)***	−0.10 (−0.11)
Tobin's q	0.99 (0.74)	0.06*** (−0.15)***	0.35 (0.12)
Number of firms	847	212	635

(a) Table 1: Summary statistics

Table 2

Collateral effects on fixed investments: basic results

This table presents the effect of collateral losses on firm investments based on least absolute distance (LAD) regressions. The dependent variable I/K is the average investment rate (defined as fixed investments over the beginning-of-period capital stock) between 1994 and 1998. $Land/K^{pre}$ is the market value of land over the replacement cost of capital in 1989. $Landco$ is a dummy variable equal to one if $Land/K^{pre}$ is above the top industry quartile and zero otherwise. Percent recent purchase is the proportion of land (in market value) purchased during 1988–1990. High Debt-to-Land is a dummy variable indicating firms with debt over market value of land in 1989 above the top quartile. I/K^{pre} is the investment rate prior to the shock in 1989. Standard errors are based on the asymptotic variance in Koenker and Bassett (1978) and are presented in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1)	(2)	(3)	(4)	(5)
Cash flow/ K	0.415*** (0.021)	0.216*** (0.024)	0.142*** (0.033)	0.126*** (0.032)	0.146*** (0.039)
Tobin's q	0.019*** (0.000)	0.005*** (0.000)	0.019*** (0.000)	0.019*** (0.000)	0.019*** (0.001)
$Land/K^{pre}$	−0.128*** (0.017)	−0.165*** (0.016)	−0.142*** (0.024)	−0.153*** (0.023)	−0.138*** (0.028)
Cash flow * $Landco$	0.569*** (0.021)	0.717*** (0.025)	0.759*** (0.034)	0.783*** (0.033)	0.761*** (0.040)
Cash/ K		0.027*** (0.004)	0.023*** (0.005)	0.027*** (0.005)	0.022*** (0.006)
Cash * $Landco$		0.102*** (0.004)	0.169*** (0.005)	0.162*** (0.005)	0.160*** (0.007)
% Recent purchase			0.002 (−0.042)	0.039 (0.041)	0.014 (0.051)
% Recent purchase * $Landco$			−0.708*** (0.109)	−0.759*** (0.105)	−0.752*** (0.126)
Leverage				0.008 (0.033)	0.01 (0.040)
$Land/K^{pre}$ * High Debt-to-Land				−0.025* (0.015)	−0.019 (0.018)
I_K^{pre}					0.019*** (0.006)
Industry dummies	Yes	Yes	Yes	Yes	Yes
Number of observations	824	751	708	708	708
Pseudo R^2	0.137	0.243	0.311	0.312	0.312

(b) Table 2: Basic investment results

Table 1 read. Table 1 compares landholding firms with the control group. Landholding firms differ from other firms in size, cash, leverage, bank-debt reliance, and Tobin's q , which motivates the later use of controls and robustness tests. The table also shows that the raw investment distribution is skewed, which helps motivate the use of LAD regressions.

Table 2 read. Table 2 contains the central investment result. The coefficient on $Land/K^{pre}$ is negative and statistically significant across specifications. The paper then adds controls for recent land purchases, leverage, high debt-to-land, and pre-shock investment. The collateral effect remains. This is the main evidence that collateral losses reduced investment.

Interpretation: Together, Tables 1 and 2 establish the basic fact pattern: landholding firms are meaningfully different, but even after controls, pre-shock collateral exposure predicts lower post-shock investment.

9.2 Table 3: Collateral effects on fixed investments: additional tests and Table 4: Summary statistics on lending relationships and top banks

Table 3

Collateral effects on fixed investments: additional tests

This table presents additional tests of the effect of collateral losses on investments. The dependent variable I/K is average fixed investments over the beginning-of-period capital stock between 1994 and 1998. $\text{Land}/K^{\text{pre}}$ is the market value of land over capital stock in 1989. Additional proxies for firm quality are measured in 1989. Landco is a dummy indicating $\text{Land}/K^{\text{pre}}$ above the top industry quartile. Percent recent purchase is the proportion of land (in market value) purchased during 1988 to 1990. High Debt-to-Land is a dummy indicating firms with debt over the market value of land in 1989 above the top quartile. $1/K^{\text{pre}}$ is the investment rate prior to the shock in 1989. Low q is a dummy variable indicating q below the median. Standard errors are based on the asymptotic variance in Koenker and Bassett (1978) and are presented in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	Group		Non-group		
	(1)	(2)	(3)	(4)	(5)
Cash flow/ K	0.120*** (0.010)	0.156*** (0.012)	0.217*** (0.041)	0.101* (0.060)	0.135*** (0.029)
Tobin's q	0.017*** (0.000)	0.017*** (0.001)	0.005*** (0.002)	0.029*** (0.000)	0.020*** (0.000)
$\text{Land}/K^{\text{pre}}$	-0.038*** (0.005)	-0.031*** (0.005)	0.030 (0.026)	-0.151*** (0.045)	-0.154*** (0.022)
Cash flow* Landco	1.442*** (0.030)	1.466*** (0.025)	-0.188*** (0.045)	0.780*** (0.062)	0.774*** (0.029)
Cash stock/ K	0.001 (0.003)	0.009** (0.004)	-0.004 (0.006)	0.030*** (0.010)	0.026*** (0.005)
Cash* Landco	0.140*** (0.005)	0.123*** (0.005)	0.000 (0.007)	0.206*** (0.011)	0.167*** (0.005)
% Recent purchase		0.053 (0.054)	0.03 (0.043)	0.019 (0.084)	0.037 (0.036)
% Recent purchase* Landco		-2.283*** (0.441)	-0.021 (0.125)	-1.716*** (0.168)	-0.619*** (0.088)
Leverage		0.117*** (0.037)	0.045 (0.037)	-0.017 (0.065)	0.02 (0.029)
$\text{Land}/K^{\text{pre}}$ High Debt-to-Land		-0.026* (0.016)	0.014 (0.016)	-0.015 (0.029)	-0.027** (0.013)
<i>Additional proxies for firm quality</i>					
Sales growth in the past three years	-0.032 (0.025)	-0.025 (0.033)			
Operating profits/Sales	0.035 (0.069)	0.072 (0.067)			
Log (assets)	0.004* (0.002)	0.003	(0.002)		
$1/K^{\text{pre}}$		0.003 (0.006)			
$\text{Land}/K^{\text{pre}}$ Low q					0.012 (0.010)
Industry dummies	Yes	Yes	Yes	Yes	Yes
Number of observations	716	651	290	418	708
Pseudo R^2	0.328	0.359	0.055	0.531	0.312

(a) Table 3: Additional investment tests

Table 4

Summary statistics on the lending relationships and the firms' top banks

This table presents descriptive statistics on banking relationships in the matched lending sample. Column (1) presents the whole-sample summary statistics averaged over the sample period between 1994 and 1998. Columns (2) and (3) present statistics for the subsamples of landholding companies and the control group. The table presents the mean of each statistic, with the median in parentheses. Landholding companies are defined as the companies with market value of land to total replacement cost of capital above the industry top quartile. The remaining firms serve as the control group. Significance levels for the difference between the landholding companies and the control groups are based on two-tailed tests; significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	The whole sample	Landholding companies	The control group
	(1)	(2)	(3)
<i>Panel A: Relationship characteristics</i>			
Duration (1984–1993)	8.32 (9.00)	8.21 (9.00)	8.37 (9.00)
Number of banks (1984–1993)	16.37 (14.00)	15.65 (13.00)	1.65 (0.00)
% Equity stake (1993)	1.61 (0.00)	1.49 (0.00)	16.63 (14.00)
Log loan growth (1994–1998 versus 1984–1989)	0.34 (0.30)	0.17*** (0.09)***	0.40 (0.42)
% of Relationships with the bigbanker	0.179	0.159	0.187
% of Relationships with the main bank	0.113	0.092*	0.122
% of firms with BBB rating or above (1989)	0.513	0.602***	0.481
% of firms with BBB rating or above (1994–1998)	0.723	0.684**	0.738
Total number of firm-bank pairs	3,194		
<i>Panel B: Top-bank characteristics</i>			
% Real estate loans	0.055 (0.037)	0.053 (0.037)	0.055 (0.037)
Capital ratio	0.03 (0.028)	0.03 (0.028)	0.03 (0.028)

(b) Table 4: Lending relationship summary

Table 3 read. Table 3 addresses alternative explanations. The paper controls for firm quality proxies such as past sales growth, operating profits over sales, size, and pre-shock investment. It also splits firms by group affiliation and investigates whether the results are driven by low- q firms. The negative collateral effect is strongest for non-group firms, consistent with relationship banks partially insulating group firms from collateral shocks.

Table 4 read. Table 4 describes the matched lending sample. It reports lending relationship duration, number of banks, equity stakes, loan growth, and top-bank characteristics. The key descriptive pattern is that landholding companies experienced lower loan growth than the control group after the shock.

Interpretation: Table 3 strengthens the investment interpretation, while Table 4 prepares the mechanism test by showing that the same collateral exposure is visible in bank lending outcomes.

9.3 Table 5: Collateral effects on corporate borrowing and Table 6: Collateral effects on fixed investments controlling for bank characteristics

Table 5

Collateral effects on corporate borrowing

This table presents the effect of collateral losses on bank lending based on Heckman two-stage regressions. The first stage is a probit regression. The dependent variable is a dummy variable indicating the survival of the lending relationship. Duration is the number of years that the firm-bank pair had positive loan balances. Big banker is a dummy indicating whether the bank served as the firm's largest lender at least once. Main bank is a dummy variable equal to one if the bank is also the firm's main bank and zero otherwise. Equity ownership is the percent of shares owned by the bank. Number of banks is the number of banks with which the firm had lending relationships. Land/ K^{1990} is the market value of land over the capital stock in 1990. Landco is a dummy variable equal to one if Land/ K^{1990} is above the top quartile in the industry and zero otherwise. Robust standard errors are presented in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1a) Probit	(1b) OLS	(2a) Probit	(2b) OLS
<i>Relationship characteristics</i>				
Duration (1984–1989)	0.193*** (0.013)	−0.421*** (0.040)	0.193*** (0.013)	−0.403*** (0.038)
Number of banks (1984–1989)	0.007 (0.078)	−0.015 (0.079)	0.017 (0.078)	−0.024 (0.078)
Bigbanker (1984–1989)	0.457*** (0.097)	−0.082 (0.083)	0.454*** (0.096)	−0.060 (0.082)
Equity ownership	0.094*** (0.021)	0.033* (0.019)	0.094*** (0.021)	0.039** (0.019)
Main bank	0.113 (0.102)	−0.120 (0.089)	0.123 (0.103)	−0.112 (0.089)
Access to bond market			0.131*** (0.050)	−0.039 (0.063)
<i>Firm characteristics</i>				
Log of assets	−0.044 (0.032)	−0.041 (0.032)	−0.039 (0.032)	−0.041 (0.032)
Tobin's q	−0.106*** (0.023)	0.001 (0.002)	−0.112*** (0.023)	0.001 (0.002)
Cash flow/ K	0.144*** (0.067)	0.103 (0.131)	0.103 (0.067)	0.103 (0.130)
Cash/ K	−0.025* (0.013)	−0.017 (0.022)	−0.023* (0.013)	−0.017 (0.021)
<i>Collateral-related variables</i>				
Land/ K^{1990}	−0.240* (0.135)	−0.487*** (0.187)	−0.294** (0.132)	−0.494*** (0.188)
% Recent purchase	4.100*** (1.050)	1.051 (0.729)	4.154*** (1.072)	1.265* (0.732)
Recentpurchase* Landco	1.026 (3.043)	−3.570 (3.387)	1.133 (3.140)	−3.507 (3.407)
Inverse Mills ratio		−1.247*** (0.472)		−1.022** (0.454)
Number of observations	3191	2626	3191	2626
Pseudo R^2/R^2	0.162	0.300	0.164	0.300

(a) Table 5: Corporate borrowing

Table 6

Collateral effects on fixed investments: controlling for bank characteristics

This table presents the effect of collateral losses on investments, controlling for the firm's top lender characteristics. The dependent variable I/K is the average fixed investments over the beginning-of-period capital stock between 1994 and 1998. Land/ K^{1990} is the market value of land over the capital stock in 1989. Landco is a dummy variable indicating Land/ K^{1990} above the top industry quartile. % Recent purchase is the proportion of land (in market value) purchased during 1988–1990. High Debt-to-Land is a dummy variable indicating firms with debt over market value of land in 1989 above the top quartile. I/K^{1989} is the investment rate prior to the shock in 1989. % Real estate loan is the bank's real estate loans over assets in 1989. Capital ratio is bank equity over assets in 1989. Standard errors are based on the asymptotic variance in Koenker and Bassett (1978) and are presented in parentheses. Significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cash flow/ K	0.191*** (0.047)	0.184*** (0.052)	0.186*** (0.047)	0.149* (0.084)	0.192** (0.092)	0.199** (0.095)	0.187* (0.109)	0.177* (0.097)
Tobin's q	0.029*** (0.000)	0.028*** (0.000)	0.036*** (0.001)	0.037** (0.017)	0.029 (0.024)	0.028 (0.021)	0.037** (0.018)	0.038* (0.020)
Land/ K^{1990}	−0.093*** (0.025)	−0.113*** (0.030)	−0.082*** (0.027)	−0.092* (0.057)	−0.114* (0.065)	−0.118* (0.064)	−0.134* (0.080)	−0.112** (0.053)
Cashflow* Landco	0.687*** (0.048)	0.760*** (0.053)	0.759*** (0.049)	0.696 (0.386)	0.764 (0.547)	0.756 (0.500)	0.745 (0.516)	0.745 (0.587)
Cash/ K	0.024*** (0.006)	0.031*** (0.007)	0.018*** (0.006)	0.028* (0.015)	0.023** (0.011)	0.024* (0.011)	0.016 (0.015)	0.023 (0.015)
Cash* Landco	0.213*** (0.006)	0.197*** (0.007)	0.196*** (0.007)	0.188** (0.088)	0.212* (0.111)	0.200** (0.098)	0.194** (0.097)	0.190** (0.082)
% Recent purchase		−0.078 (0.094)	−0.100 (0.088)	−0.045 (0.098)		−0.046 (0.119)	−0.115 (0.126)	−0.1 (0.110)
% Recent purchase* Landco		−2.496*** (0.338)	−2.509*** (0.305)	−2.417** (1.161)		−2.466 (2.224)	−2.246 (1.738)	−2.169 (2.767)
Leverage		0.052 (0.043)	0.018 (0.039)	0.007 (0.044)		0.042 (0.065)	0.007 (0.067)	0.028 (0.063)
Land/ K^{1990} * High Debt-to-Land		−0.013 (0.015)	−0.006 (0.025)	−0.014 (0.025)		−0.008 (0.020)	−0.012 (0.023)	−0.009 (0.019)
$I_{-1}K^{1990}$			0.091*** (0.008)	0.103* (0.058)			0.117* (0.063)	0.117* (0.070)
<i>Additional proxies for firm quality</i>								
Sales growth in the past three years				−0.099 (0.175)				−0.063 (0.134)
Operating profits/Sales				0.199 (0.189)				0.277 (0.203)
Log (assets)				0.007 (0.005)				0.004 (0.006)
<i>Bank characteristics</i>								
% Real estate loans	−0.286* (0.177)	−0.429** (0.190)	−0.366** (0.177)	−0.339* (0.196)				
Capital ratio	2.139** (0.984)	2.833*** (1.071)	2.263** (1.001)	2.538* (1.319)				
Bank dummies	No	No	No	No	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	472	420	420	420	472	420	420	420
Pseudo R^2	0.460	0.551	0.556	0.558	0.482	0.570	0.576	0.579

(b) Table 6: Bank controls

Table 5 read. Table 5 estimates the effect of collateral losses on bank lending using Heckman two-stage regressions. In the first stage, firms with larger pre-shock land exposure are less likely to continue lending relationships. In the second stage, conditional on continuation, they receive lower loan growth. The inclusion of bank fixed effects means the comparison is made among borrowers of the same bank.

Table 6 read. Table 6 controls for the financial health of the firm's top lender, including the lender's exposure to real estate loans and its capital ratio. The collateral effect remains. Columns with bank dummies further absorb unobserved bank characteristics. This table is central for separating the borrower collateral channel from the bank lending channel.

Interpretation: Tables 5 and 6 provide the paper's most important mechanism evidence: collateral losses reduce debt capacity, and this effect is not simply a by-product of weak banks.

10 Detailed derivation of empirical coefficients

10.1 From land holdings to investment reduction

Let the land-price collapse be a proportional decline $\rho > 0$, so that

$$\frac{\Delta P^{land}}{P^{land}} = -\rho.$$

Then firm i 's collateral loss is

$$Loss_i = \rho \cdot Land_i^{1989}.$$

If investment depends on collateral through a constrained borrowing relationship,

$$\frac{I_i}{K_i} = \alpha - \kappa \frac{Loss_i}{K_i} + X_i' \beta + \varepsilon_i,$$

then substituting the loss expression gives

$$\frac{I_i}{K_i} = \alpha - \kappa \rho \frac{Land_i^{1989}}{K_i} + X_i' \beta + \varepsilon_i.$$

Therefore the reduced-form coefficient on $Land_i^{1989}/K_i$ equals

$$\delta = -\kappa \rho.$$

Since $\rho > 0$ and $\kappa > 0$, the theory predicts $\delta < 0$.

10.2 Why the same land variable predicts lending

For lending, suppose bank j offers firm i credit according to

$$Loan_{ij} = \eta_j + \theta C_i + X_i' \beta + R_{ij}' \gamma + \varepsilon_{ij}.$$

After the land shock,

$$\Delta Loan_{ij} = \eta_j + \theta \Delta C_i + X_i' \beta + R_{ij}' \gamma + \varepsilon_{ij}.$$

Substitute $\Delta C_i = -\rho Land_i^{1989}$:

$$\Delta Loan_{ij} = \eta_j - \theta \rho Land_i^{1989} + X_i' \beta + R_{ij}' \gamma + \varepsilon_{ij}.$$

Thus, within a bank, greater pre-shock landholdings predict lower post-shock lending if banks respond to collateral losses.

10.3 Extensive and intensive lending margins

The lending channel can operate on two margins:

$$\begin{aligned} \Pr(Relationship\ survives_{ij} = 1) &= \Phi(\alpha + \delta_S Land_i^{1989}/K_i + X_i' \beta + R_{ij}' \gamma + u_j), \\ LoanGrowth_{ij} \mid Relationship\ survives_{ij} = 1 &= \alpha + \delta_G Land_i^{1989}/K_i + X_i' \beta + R_{ij}' \gamma + u_j + \varepsilon_{ij}. \end{aligned}$$

The prediction is $\delta_S < 0$ and $\delta_G < 0$: collateral losses lower the probability of relationship continuation and reduce loan growth conditional on continuation.

11 How the paper rules out alternative explanations

11.1 Firm quality and investment opportunities

A major concern is that firms with more land may simply be low-quality firms. The paper addresses this by controlling for Tobin's q , cash flow, cash stock, pre-shock investment, past sales growth, profitability, and size. The collateral coefficient remains negative.

11.2 Overinvestment correction

Another possible interpretation is that landholding firms overinvested during the bubble, so the post-bubble decline is merely a correction. The paper tests whether the effect is stronger among low- q firms. The evidence does not support the overinvestment explanation. The paper also finds evidence that collateral affects major investments, which is more consistent with underinvestment caused by financing constraints.

11.3 Keiretsu and main bank relationships

The collateral effect is weaker or absent for group-affiliated firms in one classification. This supports the idea that banking relationships can partially insure firms against collateral shocks. However, the result is not fully robust to alternative group classifications, suggesting that the main-bank system is not cleanly measured.

11.4 Bank balance-sheet damage

Bank financial condition matters, but it does not explain away the collateral channel. Even after controlling for lender real estate exposure, lender capital ratio, and bank fixed effects, firm collateral losses continue to predict lower investment.

12 Economic intuition

12.1 Why collateral matters even with strong banking relationships

One might expect Japanese relationship banks to smooth credit through temporary collateral shocks. The paper shows that they do not fully do so. Relationship lending mitigates but does not eliminate the role of collateral, because collateral affects both formal debt capacity and the lender's downside protection.

12.2 Why real estate is a powerful collateral asset

Real estate is durable, observable, pledgeable, and widely accepted by banks. Therefore, movements in real estate prices directly change borrowing capacity. This makes property bubbles especially relevant for real activity, because a fall in property prices can reduce investment through credit constraints even for non-real-estate firms.

12.3 Why this is a financial accelerator mechanism

The results fit the financial accelerator logic. Lower collateral value reduces borrowing. Lower borrowing reduces investment. Lower investment reduces future cash flow and possibly future collateral value. This feedback can amplify an initial asset-price shock into a broader real downturn.

13 Limitations

13.1 External validity

Japan's land-price collapse was unusually large. The direction of the collateral channel is likely general, but the magnitude should not be mechanically extrapolated to ordinary real estate cycles.

13.2 Public firms and manufacturing firms

The sample consists of publicly traded manufacturing firms. Smaller private firms may be more financially constrained, so the collateral channel could be even stronger for them. On the other hand, their accounting and collateral data may be harder to measure.

13.3 Measurement of land value

The market value of land is estimated using a perpetual inventory method. The paper performs robustness checks with regional land-price variation, but the exact location and quality of every parcel are not observed.

14 Connection to broader literature

14.1 Credit cycles and financial accelerator

The paper provides micro evidence for the credit-cycle mechanism in Kiyotaki and Moore (1997). It shows that collateral values affect debt capacity and investment, not merely in theory but in matched firm-bank data.

14.2 Corporate investment and financing constraints

The paper complements Fazzari, Hubbard, and Petersen (1988) by offering an alternative measure of financing frictions: collateral rather than internal cash flow. This is valuable because investment-cash flow sensitivity is difficult to interpret.

14.3 Asset bubbles and real activity

The paper links asset-price bubbles to real investment through balance sheets. It shows that the bursting of an asset bubble can reduce investment even outside the asset-producing sector because firms use assets as collateral for borrowing.

15 Conclusion and takeaway

15.1 Core conclusion

Gan (2007) shows that collateral value has a large and causal effect on corporate investment. The Japanese land-price collapse reduced firms' pledgeable collateral. Firms with larger pre-shock landholdings lost more debt capacity, received less bank credit, and invested less.

15.2 Key takeaway for research

The paper is valuable because it combines a natural experiment with mechanism evidence. The investment result alone would be suggestive; the matched bank-firm lending evidence makes the collateral-channel interpretation much more convincing.

15.3 Key takeaway for policy

Property-market collapses can affect real activity through the balance sheets of nonfinancial firms. Policies that focus only on banks may miss the borrower-side collateral channel. Stabilizing credit markets after asset-price collapses requires attention to both lender health and borrower collateral capacity.